

The Prior Is Not the Bottleneck: Where Success Comes From in Guided Diffusion Motion Planning

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Abstract

Guided diffusion is a popular paradigm for robot motion planning, and a steady stream of work improves the *generative prior*: better noise schedules, richer trajectory representations, and environment conditioning. We present a controlled study on Guided Polynomial Diffusion (GPD), re-implemented in the EDMP codebase, showing that for a guided planner **the prior is largely not the bottleneck**. Across five independent prior-side interventions—recalibrating the noise schedule, increasing polynomial capacity, adding a smoothness penalty, conditioning the denoiser on the scene with classifier-free guidance, and training $5\times$ longer (to 1M steps)—planning success rate (SR) never improves, with deltas ranging from statistically zero to small significant *decreases*, even when the interventions lower training loss or reconstruction error. A failure-mode analysis isolates *continuous (swept)* infeasibility as the dominant residual failure. Acting on it, two interventions on the *feasibility machinery*—faithful-collision RRT stitching (+14 *pp*) and a trajectory-optimization repair (+8.9 *pp*, 95% CI [7.6, 10.2])—break the plateau that no prior-side change could. A naive-seed control sharpens the claim: the prior is *necessary* as an initializer (a straight-line seed with the same repair reaches only 55% vs. 82%) but *improving* it does not help. The repair lever transfers to a second planner (EDMP, +10.3 *pp*) and collapses the diagnosed swept-collision failure (15.5% $\rightarrow$ 3.0%). We argue the community’s effort is mis-allocated and release an open, GPU-native re-implementation with all ablations.

1 Introduction

Diffusion models generate diverse, multimodal trajectories and, with classifier (cost-gradient) guidance, can be steered to avoid obstacles at inference time [7, 3, 9]. GPD [1] compresses a trajectory into eight Bernstein control points and shortens the diffusion chain to $T=64$ for fast planning. Most follow-up work—and the natural set of “obvious” improvements—targets the generative model: the noise/SNR schedule, the trajectory parameterization, and conditioning the denoiser on the environment.

We ran those improvements as controlled, single-variable experiments and found they **do not raise success rate** (Fig. 1). This paper reports that negative result, explains it, and then shows where the gains actually are: the inference-time feasibility machinery. Our central, somewhat contrarian claim is that for guided diffusion planners a competent prior is “good enough,” and the returns lie in collision-aware stitching and trajectory-optimization repair.

The stakes are practical. Retraining a prior—a new schedule, a larger basis, scene conditioning, or simply a longer run—costs GPU-days and is the default first move when a diffusion planner underperforms. If, as we find, that move is systematically ineffective while a retraining-free repair step yields double-digit gains, then the field’s effort is mis-directed. We make this case not with a single comparison but with five prior-side interventions and two feasibility levers, each measured under one continuous-execution success metric with confidence intervals, so the asymmetry in Fig. 1 cannot be attributed to noise or to a single favorable configuration.

Contributions.

1. A **decoupling result**: five prior-side improvements (including a $5\times/1\text{M}$ -step training budget) each improve a nominal generative metric yet never raise SR—each backed by a per-intervention confidence interval.
2. A **failure-mode diagnosis** isolating continuous/swept infeasibility as the dominant residual failure, and its *closure* after repair.
3. **Two positive levers on the feasibility machinery**: faithful-collision stitching (+14 *pp*) and trajectory-optimization repair (+8.9 *pp* on GPD, +10.3 *pp* on EDMP—planner-agnostic).
4. A **naive-seed control** scoping the claim: the prior is a necessary initializer, not a tunable lever.
5. An **open, GPU-native** re-implementation with every ablation.

2 Related Work

Diffusion for motion planning. Diffuser [7] framed trajectory generation as denoising; MPD [3] and EDMP [9] add cost-gradient (classifier) guidance to enforce collision avoidance at inference, EDMP using an ensemble of guidance costs. GPD [1] makes this fast by compressing trajectories to a low-degree Bernstein basis and shortening the diffusion chain. This line of work overwhelmingly improves the *prior*: the schedule, the representation, or scene conditioning. We instead hold the prior fixed and interrogate whether those improvements pay off.

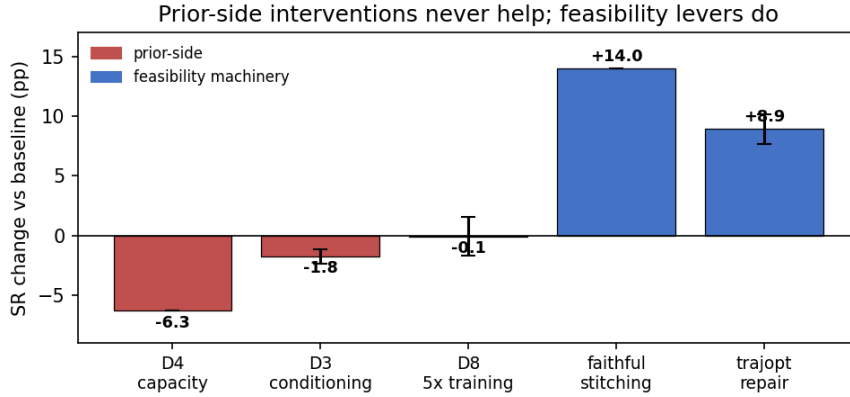


Figure 1: Success-rate change for each intervention on the hybrid benchmark. Every prior-side change (red) is zero or negative; both feasibility-machinery levers (blue) are large and positive. Error bars are 95% confidence intervals where multi-seed runs exist. This asymmetry—prior improvements never help, feasibility repair does—is the paper’s central finding.

Feasibility repair. A parallel thread refines a learned or sampled seed with trajectory optimization: PRESTO [2] and MPD [3] warm-start an optimizer from a diffusion sample. We adopt this mechanism but use it as an *analytical instrument*—contrasting it against five prior-side changes under one metric—rather than as a standalone method.

Learned and classical baselines. MPiNets [4] regresses policies from large demonstration sets, and cuRobo [10] is a GPU-parallel classical planner; both define the strong end of the speed/quality frontier we position against qualitatively (our absolute SR is below published systems; §3).

3 Background and Setup

GPD. A TemporalUNet ϵ -predictor (~ 3.95 M params) over eight Bernstein control points, $T=64$, variance schedule $\beta = \text{linspace}(0, 0.02, T+1)$; start/goal are pinned at the boundary control points. Inference samples K candidates under cost-gradient guidance (differentiable AABB intersection-volume plus swept-volume via PyTorch forward kinematics), then selects and stitches them [1, 5].

Success metric. We execute the trajectory in PyBullet under position control; success requires *no obstacle contact during execution*. This is a continuous-execution test, not a waypoint check.

Setup. Unless noted, numbers are on the MPiNets [4] *hybrid*-solvable set (Fig. 2), balanced 100 scenes per environment type (400 total), under the GPD-stitched configuration (single guide, $K=32$, faithful-collision RRT-Connect stitching [8]). Our re-implementation reproduces the method ordering of [1] but sits ~ 21 pp below its absolute numbers; all claims below are therefore *deltas measured within our own system*, never cross-method absolutes.

4 The Prior Is Not the Bottleneck

We test the five most natural prior-side improvements. Each is a clean single-variable change sharing an identical guidance/stitching pipeline.

4.1 Noise-schedule recalibration

At $T=64$ the linear schedule gives $\bar{\alpha}_T \approx 0.52$ —the model is never trained near pure noise, while inference starts from $\mathcal{N}(0, I)$. The “fix” (raise terminal noise, or use cosine) *lowered* SR on all scene types despite a *lower* training loss ($0.53 \rightarrow 0.43$). The low-terminal-noise regime keeps the reverse process near-identity, acting as an implicit smoothness prior; injecting variance yields less-smooth, more-colliding paths. ϵ -loss is a poor proxy for planning SR.

4.2 Polynomial capacity

Bernstein reconstruction error shows degree-7 ($n=8$) discards $5\text{--}17^\circ$ of joint motion on the constrained demonstrations (mean 5.47°), so the representation *is* lossy where GPD struggles. Yet training $n=16$ ($4\times$ lower reconstruction error, matched convergence loss) yields a **net** -6.3 pp: it beats $n=8$ on sharp-turn dresser scenes but loses on tabletop/cubby, and costs $5\times$ the training.

Mechanism (compression = smoothness regularizer). Measuring the raw sampled trajectories (100 scenes, before any repair), the compact $n=8$ prior is markedly smoother: joint-jerk RMS 0.00166 vs. 0.00214 (-22% ; Fig. 3), with a larger clearance margin. The low-degree basis cannot express the high-frequency motion a higher-capacity model fits, so it acts as an implicit smoothness prior—and smoother trajectories collide less under continuous execution. Reconstruction fidelity is *anti-correlated* with the smoothness that drives SR.

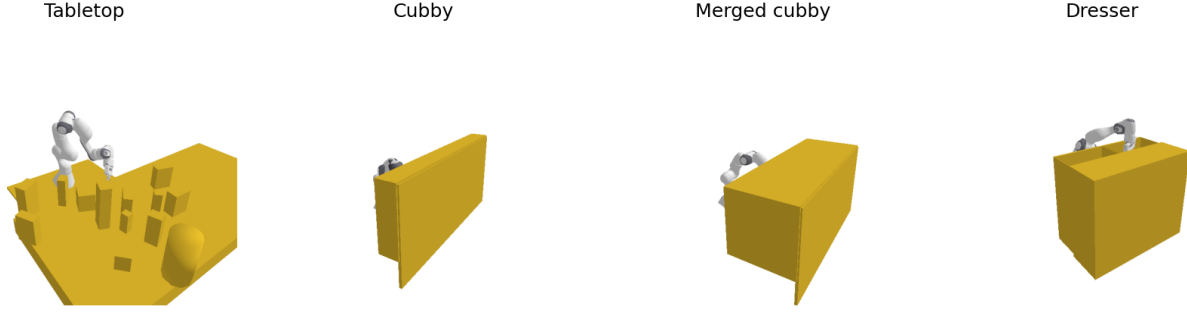


Figure 2: The four MPiNets [4] environment types, with the Franka Panda at a goal configuration reaching into each scene (obstacles in yellow). Tabletop scenes scatter small objects; cubby and merged-cubby require reaching through shelf openings; dresser requires reaching into a box—the constrained types where swept infeasibility dominates.

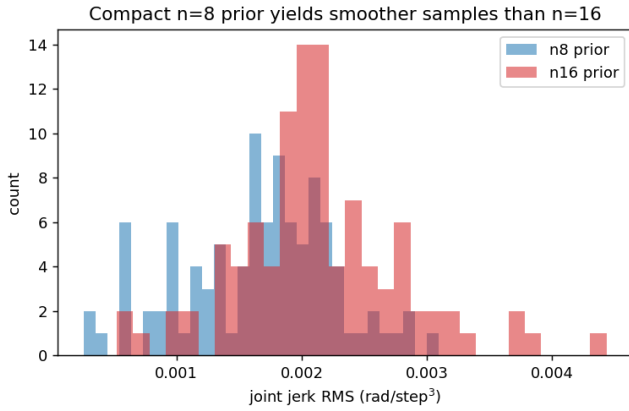


Figure 3: Distribution of raw-sample joint jerk over 100 hybrid scenes. The compact $n=8$ prior is smoother than the higher-capacity $n=16$ prior, a direct mechanism for why capacity hurts SR despite lower reconstruction error.

4.3 Scene-conditioned denoiser (CFG)

We add a masked-DeepSets [11] obstacle encoder whose embedding feeds the time embedding, with a learned null embedding for classifier-free guidance [6]. The conditioned model reaches a *lower* loss ($0.502 < 0.52$), yet with guidance removed the conditioned prior plans no better than the scene-blind prior: scene information enters through *guidance*, not the prior. Over five paired seeds the conditioning delta is -1.75 pp (95% CI $[-2.37, -1.13]$)—a small but significant *decrease*.

4.4 Training to convergence ($5\times$, 1M steps)

The most basic prior-side lever is more training. We train the canonical $n=8$ prior for 1,000,000 steps ($5\times$ the baseline); ϵ -loss drops $\sim 4\%$ and flattens for the final 300k steps. Over five paired seeds the delta is -0.10 pp (95% CI $[-1.71, +1.51]$)—statistically zero. This rules out under-training as the explanation for both the flat prior-side results and the reproduction gap.

4.5 Summary

Five prior improvements, five nominal-metric gains, **no positive SR gain**: deltas span statistically zero ($5\times$ training) to small significant decreases (conditioning -1.75 pp; schedule and capacity larger). Against a measured baseline sampling noise $\sigma \approx 1.6$ pp, no prior intervention clears it upward, while the repair lever below clears it by $\sim 6\sigma$.

5 Where Success Actually Comes From

5.1 Failure-mode analysis, and its closure

We categorize 200 hybrid failures of the baseline, and re-categorize after repair:

failure mode	baseline %	after repair %
success	74.0	78.5
goal/start IK infeasible	2.5	1.5
path_collision (waypoints)	8.0	17.0
dynamic_only (swept)	15.5	3.0

The executor and static checker share a contact model, so `dynamic_only` is genuine: the robot *sweeps through obstacles between waypoints* (Fig. 4). It is the dominant residual failure, and the repair we build (§5.3) collapses it to 3.0%. Of the eliminated swept failures, ~ 4.5 pp convert to success and the rest become statically-detectable `path_collision`: on the hardest scenes the optimizer churns but leaves a residual static collision—it converts dynamic failures, it does not manufacture them.

5.2 Lever 1: faithful-collision stitching

The stitcher originally certified bridges with the loose AABB intersection-volume *proxy*, marking colliding bridges as free. Replacing it with a faithful PyBullet check inside RRT-Connect yields $+14$ pp over the proxy plateau, the first intervention to move SR.

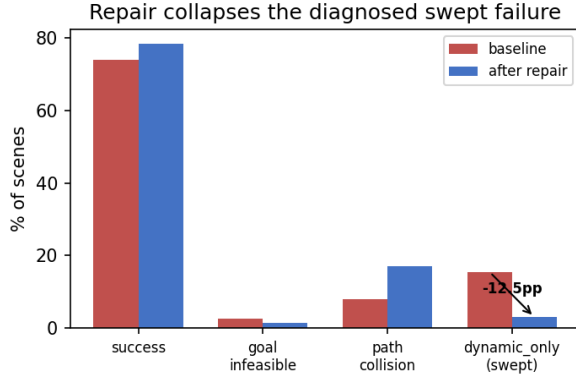


Figure 4: Failure-mode distribution before and after repair. The dominant baseline failure—*dynamic_only* swept collision—collapses from 15.5% to 3.0%, the very failure the repair objective (Eq. 1) targets.

5.3 Lever 2: trajectory-optimization repair

Motivated by the analysis above and recent work [2, 3], we add a post-hoc optimizer. With endpoints q_0, q_N pinned to start/goal, we solve over the interior waypoints $q_{1:N-1}$

$$\min_{q_{1:N-1}} \sum_i c(q_i) + \lambda_m \sum_i c\left(\frac{q_i + q_{i+1}}{2}\right) + \lambda_s \sum_i \|q_{i-1} - 2q_i + q_{i+1}\|^2, \quad (1)$$

where $c(\cdot)$ is the guide’s differentiable collision cost. The first term drives waypoints out of collision; the second penalizes collision at *edge-midpoints*, directly targeting the swept infeasibility diagnosed above; the third is a smoothness (acceleration) penalty. We run Adam with joint-limit projection and *early-stop* on a densified faithful collision check, returning a densified trajectory for overshoot-free execution. It reuses the existing differentiable cost—no retraining.

Over five seeds, repair raises SR from 72.9 ± 1.6 to 81.8 ± 1.3 : a paired $+8.9pp$, **95% CI** [7.6, 10.2], $t(4)=19.4$, $p<0.0001$. The optimizer targets a densified static check; the benchmark independently verifies dynamic execution, so this is a real executable-SR gain at $\sim +1.8s$ /scene (only on scenes needing repair).

5.4 Is the prior even needed? A naive-seed control

If improving the prior never helps, is the prior irrelevant? We replace the diffusion candidates with a single straight-line joint-space seed and run the *identical* downstream (RRT stitch $\pm$ repair):

seed	stitch only	stitch + repair
linear (no prior)	18.5 ± 0.3	55.0 ± 1.0
diffusion (prior)	72.9 ± 1.6	81.8 ± 1.3

The prior is **necessary as an initializer**: even the best repair on a linear seed reaches only 55%, $-26.8pp$ below

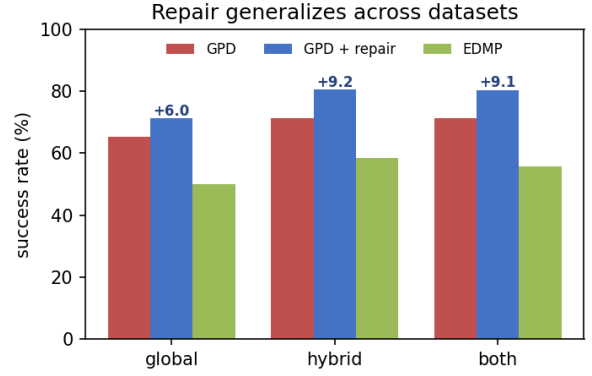


Figure 5: Repair generalizes across all three MPiNets solvable splits (+6 to +9 *pp*), and GPD+repair beats EDMP everywhere at a fraction of its planning time (Table below).

diffusion+repair—repair cannot substitute for a competent seed. Yet repair does far more *work* on the bad seed (+36.5 *pp*) than on the diffusion seed (+8.9 *pp*): good initialization and good repair are complementary, not interchangeable. This scopes the paper’s claim precisely—*improving* an already-competent prior buys nothing, but the prior is not dispensable.

5.5 The repair lever generalizes (EDMP)

If the bottleneck is the feasibility machinery, the same repair should help a different planner. Applied unchanged to EDMP [9] (a different model: $T=255$ over full 50-waypoint trajectories), repair lifts SR from 57.2% to 67.5% (+10.3 *pp*; cubby +18, merged +12, dresser +10)—*larger* than the GPD gain and on the same constrained scene types. The lever is planner-agnostic.

5.6 Generalization across datasets

The repair lever is not a hybrid-only artifact (Fig. 5). On the three MPiNets solvable splits (600 natural-weighted scenes each), it lifts SR by +6 to +9 *pp* everywhere, and consistently beats EDMP at a fraction of its planning time:

split	GPD	GPD+repair	EDMP
global	65.2	71.2 (+6.0)	50.0
hybrid	71.3	80.5 (+9.2)	58.5
both	71.2	80.3 (+9.1)	55.7
avg time	7.6s	9.4s	47.6s

The gain is largest on the constrained (hybrid/both) splits where swept infeasibility dominates, consistent with the failure-mode diagnosis.

6 Discussion

The two effective levers are both on the *inference-time feasibility machinery*, and both concern *continuous* feasibility

(collision-free bridges; collision-free swept motion). The ineffective levers are all on the prior, and several improved nominal generative metrics. A recurring theme is that *fidelity is decoupled from success at every level*: ϵ -loss (priors), reconstruction error (capacity), and even the differentiable repair *objective* (an exact sphere-SDF objective ties the cheap AABB proxy; App. A.2), because the faithful check is the true oracle. For guided diffusion planners, effort spent improving priors yields little once a competent guided prior exists; the returns are in feasibility repair.

When would the prior matter? Our claim is scoped, not absolute. The naive-seed control (§5.4) shows the prior is indispensable as an initializer—remove it and repair stalls at 55%—so a prior good enough to place the seed in roughly the right homotopy class is a prerequisite, not a target for further tuning. We expect prior improvements to pay off in regimes our benchmark does not stress: much higher-DoF systems, tasks with hard end-effector constraints the guidance cost cannot express, or settings without a fast faithful collision checker to serve as the repair oracle. Absent those, the empirical picture is stark—five prior-side changes, no gain; two feasibility changes, double-digit gains.

Limitations. Core results are on hybrid-solvable, balanced 400; we generalize the repair lever to global/both splits and to EDM, but larger N and more planners would tighten the claim. Absolute SR is below the published GPD system; we therefore report deltas within a fixed system rather than cross-method absolutes.

7 Conclusion

In a controlled study of guided polynomial diffusion planning, the generative prior is not the bottleneck: five natural prior improvements never raise SR while improving nominal metrics, and a $5\times$ prior changes nothing. The gains live in the feasibility machinery—faithful-collision stitching (+14 *pp*) and trajectory-optimization repair (+8.9 *pp* on GPD, +10.3 *pp* on EDM)—which also collapses the diagnosed swept-collision failure. The prior earns its place as a necessary initializer, not as a tunable lever. We release the implementation and all ablations to support feasibility-centric research on diffusion planners.

A Ablations

A.1 Trajopt hyperparameters (time/accuracy Pareto)

One-factor-at-a-time around the default (iters 60, densify 3, midpoint-weight 1.0), hybrid 400 (Fig. 6). *Iterations peak, not plateau*: SR saturates by ~ 40 and iters 100 regresses (-3.2 *pp*)—over-optimization drifts waypoints off the smooth seed. The *midpoint* term is causal for swept feasibility (-3.4 *pp* without it). Output densification peaks at 3.

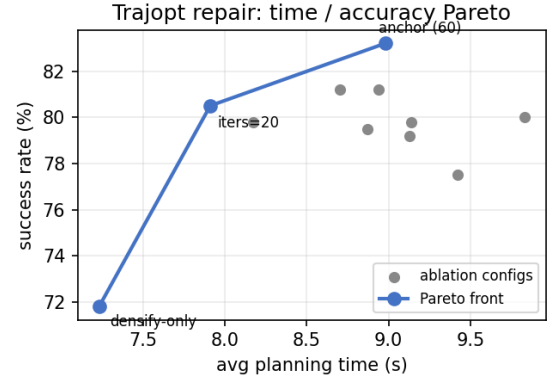


Figure 6: Time/accuracy trade-off of the trajopt-repair configurations. The Pareto front runs densify-only $\rightarrow$ iters 20 $\rightarrow$ the anchor (iters 60); all other settings are dominated.

knob	value	SR%	avg t (s)
iters	0 (densify-only)	71.8	7.2
	20	80.5	7.9
	40	81.2	8.7
	60	83.2	9.0
	100	80.0	9.8
out-densify	1	77.5	9.4
	3	83.2	9.0
	5	79.5	8.9
mid-weight	0.0	79.8	8.2
	1.0	83.2	9.0
	2.0	81.2	8.9

A.2 Exact vs. proxy collision objective

We implemented an exact GPU-batched sphere-SDF objective (FK-placed link spheres vs. oriented obstacle boxes), removing the AABB double-inflation. Swapped into the same repair loop (3 seeds) it gives 82.0 ± 0.9 vs. 81.8 ± 1.3 for the proxy—a paired +0.6 *pp*, not significant, at equal time. Because the faithful PyBullet check is the stopping oracle, objective fidelity is not a lever.

A.3 Per-intervention seed CIs

Five paired seeds, hybrid 400. Baseline 72.9 ± 1.6 ; conditioning 71.2 ± 1.3 ($\Delta = -1.75$, CI $[-2.37, -1.13]$); $5\times$ training 72.8 ± 0.9 ($\Delta = -0.10$, CI $[-1.71, +1.51]$); repair 81.8 ± 1.3 ($\Delta = +8.9$, CI $[7.6, 10.2]$).

References

- [1] Anonymous. Guided polynomial diffusion for motion planning. *arXiv preprint arXiv:2501.18229*, 2025.
- [2] Anonymous. Presto: Fast motion planning using diffusion models based on key-configuration environment representation. *IEEE*

International Conference on Robotics and Automation (ICRA), 2025.

- [3] Joao Carvalho, An T. Le, Mark Baierl, Dorothea Koert, and Jan Peters. Motion planning diffusion: Learning and planning of robot motions with diffusion models. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.
- [4] Adam Fishman, Adithyavairavan Murali, Clemens Eppner, Bryan Peele, Byron Boots, and Dieter Fox. Motion policy networks. In *Conference on Robot Learning (CoRL)*, 2022.
- [5] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [6] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022.
- [7] Michael Janner, Yilun Du, Joshua B. Tenenbaum, and Sergey Levine. Planning with diffusion for flexible behavior synthesis. *International Conference on Machine Learning (ICML)*, 2022.
- [8] James J. Kuffner and Steven M. LaValle. Rrt-connect: An efficient approach to single-query path planning. *IEEE International Conference on Robotics and Automation (ICRA)*, 2000.
- [9] Kallol Saha et al. Edmp: Ensemble-of-costs-guided diffusion for motion planning. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.
- [10] Balakumar Sundaralingam et al. curobo: Parallelized collision-free robot motion generation. *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.
- [11] Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabas Poczos, Ruslan Salakhutdinov, and Alexander Smola. Deep sets. *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.